



Hospital Engineering and Management

Lecture 12: Hospital Electrical Systems and Essential/Emergency Power Supply

Dr. Sahaj Saxena

Department of Electrical and Instrumentation Engineering
TIET

Course Code: UBM702

Learning Objectives

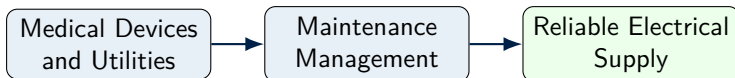
By the end of this lecture, students should be able to:

- ▶ explain the architecture of a hospital electrical distribution system;
- ▶ classify hospital loads according to clinical criticality and continuity requirements;
- ▶ describe transformers, switchgear, generators, UPS, ATS and battery backup;
- ▶ explain earthing, bonding, isolation, leakage-current and shock hazards in medical locations;
- ▶ perform basic load, generator and UPS sizing calculations;
- ▶ identify electrical-safety, power-quality and maintenance requirements in hospitals;
- ▶ relate electrical-system design decisions to patient safety and biomedical equipment reliability.

Connection with Lecture 11

Lecture 11 Established

Maintenance management keeps medical equipment and hospital utilities available, safe and traceable throughout their life cycle.



Central Idea

A perfectly maintained ventilator is still useless if safe electrical power is unavailable when the patient needs it.

Opening Scenario: Power Fails During Surgery

A patient is undergoing surgery. Suddenly the normal utility supply fails.

What Must Not Stop?

- ▶ anaesthesia workstation;
- ▶ patient monitoring;
- ▶ selected surgical equipment;
- ▶ essential lighting;
- ▶ medical-gas control and alarms;
- ▶ communication systems.

Engineering Response

- ▶ alternate power source;
- ▶ automatic transfer;
- ▶ no-break UPS where required;
- ▶ battery-backed alarms;
- ▶ selective load restoration;
- ▶ trained outage response.

In a hospital, continuity of power can become continuity of life.

Why Hospital Electrical Systems Are Special

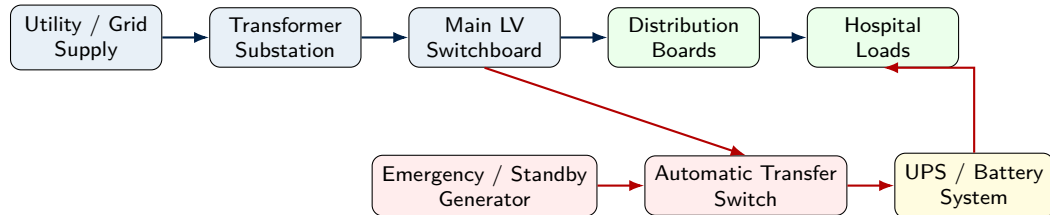
Hospital electrical systems differ from ordinary commercial buildings because they support:

- ▶ life-support and physiologic-monitoring equipment;
- ▶ invasive procedures where patients are electrically vulnerable;
- ▶ critical diagnostic and therapeutic technology;
- ▶ operating theatres, ICUs, emergency departments and laboratories;
- ▶ medical gases, HVAC, water, communication and IT systems;
- ▶ fire detection, emergency lighting and evacuation systems;
- ▶ services that must operate continuously, 24 hours a day.

Design Objective

Provide electrical power that is **safe, reliable, maintainable, selective, resilient and appropriate to clinical risk.**

Hospital Electrical System: Big Picture



Key Principle

Normal and emergency sources are coordinated so that critical loads continue with an interruption duration appropriate to their clinical risk.

From Utility to Bedside

A typical distribution chain contains:

1. incoming utility supply;
2. high-voltage / medium-voltage switchgear where applicable;
3. transformers;
4. main low-voltage switchboard;
5. busbars and rising mains;
6. floor or departmental distribution boards;
7. final circuits and socket outlets;
8. local UPS, isolation or equipment-specific power conditioning where required.

Engineering Lesson

Reliability depends on the **entire chain**. A single weak switchboard, cable, breaker or connector can interrupt a clinically critical service.

Load Classification by Clinical Consequence

For teaching, hospital loads can be grouped by the consequence of losing power.

No-Break / Near No-Break

Interruption may immediately affect patient safety or data integrity.

- ▶ selected life support;
- ▶ critical monitoring;
- ▶ selected IT/network loads.

Short-Break Tolerable

Brief interruption may be acceptable if alternate power is rapidly restored.

- ▶ selected essential lighting;
- ▶ selected critical outlets;
- ▶ some pumps / support systems.

Non-Critical / Deferrable

Temporary loss does not immediately endanger the patient.

- ▶ offices;
- ▶ nonessential comfort loads;
- ▶ selected general services.

Exact categories and transfer times must follow the code adopted by the institution and its risk assessment.

Criticality Is a Clinical Question First

Consider two identical socket outlets:

- ▶ one powers a television in a waiting area;
- ▶ the other powers an infusion pump in an ICU.

The electrical load may be similar, but the **clinical consequence of failure is completely different**.

Risk-Based Design

Clinical consequence → required continuity → redundancy → backup source → monitoring and maintenance

Important

Electrical-system design should not be based only on kW or kVA. It must also consider **what the load does to the patient**.

Connected Load, Maximum Demand and Diversity

- ▶ **Connected load:** sum of rated loads connected to the system.
- ▶ **Maximum demand:** highest expected simultaneous demand.
- ▶ **Demand factor:** maximum demand divided by connected load.
- ▶ **Diversity:** not all loads operate at full rating at the same time.
- ▶ **Future margin:** allowance for expansion and new medical technology.

Illustrative Relation

$$P_{design} = P_{connected} \times f_{demand} \times (1 + f_{future})$$

Never use a generic demand factor without a load study and applicable design standard.

Numerical Example: Estimate Design Demand

Suppose a hospital block has:

$$P_{connected} = 1000 \text{ kW}, \quad f_{demand} = 0.70$$

and a 20% future-growth allowance.

$$P_{design} = 1000 \times 0.70 \times 1.20 = 840 \text{ kW}$$

If the expected power factor is 0.90:

$$S = \frac{840}{0.90} = 933.3 \text{ kVA}$$

Interpretation

The calculated value is a planning estimate. Final transformer, generator and feeder selection must also consider redundancy, motor starting, harmonics, ambient conditions, protection and code requirements.

Transformer Substation

The transformer substation converts incoming supply to the voltage level used by the hospital distribution system.

- ▶ transformer capacity must match demand and future expansion;
- ▶ redundancy may be required for high-criticality facilities;
- ▶ ventilation and heat dissipation are important;
- ▶ protection must coordinate with upstream and downstream devices;
- ▶ access, fire safety and maintenance clearance must be planned;
- ▶ transformer losses affect energy efficiency and operating cost.

Biomedical Engineering Connection

Voltage quality at the substation ultimately influences the reliability of sensitive imaging, laboratory, monitoring and life-support equipment.

Main Switchboard and Distribution Boards

Major functions include:

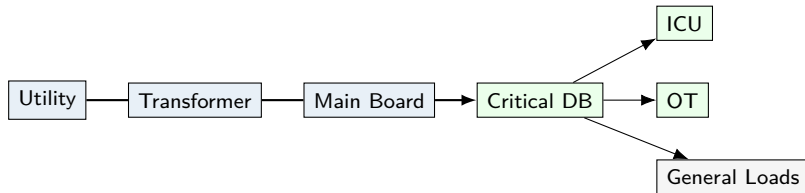
- ▶ receiving and sectionalizing electrical power;
- ▶ switching and isolation;
- ▶ overcurrent and short-circuit protection;
- ▶ metering and energy monitoring;
- ▶ feeding normal and essential distribution networks;
- ▶ supporting safe maintenance and fault localization.

Design Concern

A fault in one area should not unnecessarily shut down unrelated critical departments. This is the objective of **selective coordination** and appropriate system segmentation.

Single-Line Diagram (SLD)

An SLD is the engineering map of the hospital electrical system.

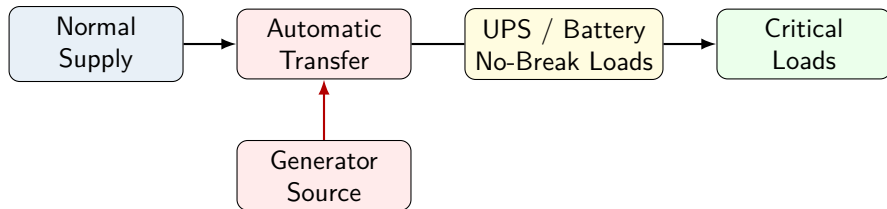


An SLD Should Help Engineers Answer

Where can power be isolated? What protects each feeder? Which source supplies a critical board? What happens if a component fails?

Essential / Emergency Power Architecture

A resilient hospital commonly combines different backup technologies because they solve different problems.



Concept

The generator provides longer-duration standby energy; the UPS bridges interruption and conditions power for selected loads.

Standby Generator System

A hospital generator system typically includes:

- ▶ engine-generator set;
- ▶ starting batteries and charger;
- ▶ fuel storage and fuel-transfer system;
- ▶ cooling and ventilation;
- ▶ exhaust system;
- ▶ generator control panel;
- ▶ synchronizing system where multiple sets operate together;
- ▶ automatic transfer and monitoring;
- ▶ routine testing and load exercise.

Common Mistake

A generator that starts successfully without meaningful load testing can still fail when a real outage occurs.

Generator Sizing: What Must Be Considered?

Generator capacity cannot be selected only from steady-state kW.

- ▶ essential connected and diversified load;
- ▶ starting current of motors and compressors;
- ▶ elevators, pumps and HVAC restart sequence;
- ▶ UPS rectifier and battery-recharge demand;
- ▶ nonlinear loads and harmonics;
- ▶ future expansion;
- ▶ altitude and ambient-temperature derating;
- ▶ redundancy philosophy such as $N+1$;
- ▶ permissible voltage and frequency dip during load steps.

Numerical Example: Generator Planning

Assume the emergency-load study gives:

- ▶ steady critical demand = 360 kW;
- ▶ future / operational margin = 25%;
- ▶ target operating power factor = 0.8.

$$P_{plan} = 360 \times 1.25 = 450 \text{ kW}$$

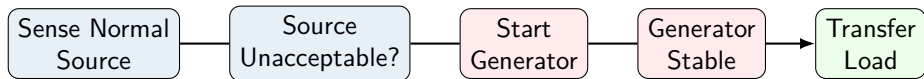
$$S_{plan} = \frac{450}{0.8} = 562.5 \text{ kVA}$$

Important

A nominal rating above 562.5 kVA may still be necessary after checking motor starting, transient response, harmonics, site derating and redundancy. This example is for concept demonstration only.

Automatic Transfer Switch (ATS)

The ATS supervises the preferred source and transfers selected loads when it becomes unacceptable.



- ▶ undervoltage / frequency sensing;
- ▶ intentional time delays to avoid nuisance transfer;
- ▶ interlocking to prevent unsafe source connection;
- ▶ retransfer logic when normal supply returns;
- ▶ testing and maintenance without compromising safety.

UPS: Why Is a Generator Not Enough?

A generator usually requires time to detect failure, start, stabilize and accept load.

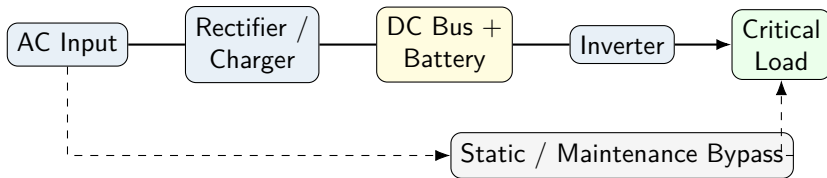
UPS Functions

- ▶ provides continuity during source transfer;
- ▶ supports no-break or near-no-break loads;
- ▶ can improve voltage and frequency quality;
- ▶ protects selected IT and sensitive electronic systems;
- ▶ provides orderly shutdown where long backup is unnecessary.

But Remember

UPS energy is finite. Battery autonomy must be matched to generator start/transfer time, clinical risk and the intended contingency.

Online UPS: Simplified Energy Path



Engineering Question

What happens if the UPS itself fails? A safe design considers bypass, redundancy, alarms and maintainability.

Numerical Example: UPS Capacity

Suppose selected ICU and IT loads total 30 kW and operate at power factor 0.90.

$$S = \frac{30}{0.90} = 33.3 \text{ kVA}$$

Allow 25% spare capacity:

$$S_{plan} = 33.3 \times 1.25 = 41.6 \text{ kVA}$$

An engineer may therefore investigate a standard size around 50 kVA, **subject to** manufacturer recommendations, transient load, redundancy and battery requirements.

Do Not Confuse

UPS kVA rating answers “how much load?” Battery capacity answers “for how long?”

Numerical Example: Battery Energy

A UPS must support a 20 kW critical load for 15 minutes.

$$E_{ideal} = 20 \times \frac{15}{60} = 5 \text{ kWh}$$

If an illustrative overall discharge/conversion factor of 0.85 is used:

$$E_{battery} \approx \frac{5}{0.85} = 5.88 \text{ kWh}$$

Real Design Also Considers

DC bus voltage, allowable depth of discharge, discharge rate, battery aging, temperature, inverter efficiency, end-of-life margin and manufacturer discharge curves.

Battery Systems

Common technologies include lead-acid and lithium-based systems, selected according to application and institutional policy.

Key engineering issues:

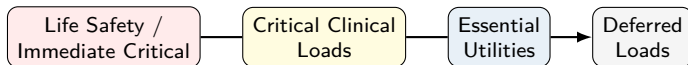
- ▶ capacity and autonomy;
- ▶ battery age and state of health;
- ▶ ambient temperature;
- ▶ ventilation and thermal management;
- ▶ charger performance;
- ▶ terminal condition and connection integrity;
- ▶ fire and chemical hazards;
- ▶ monitoring and replacement planning.

Maintenance Principle

Battery failure is often silent until backup power is actually demanded; periodic testing is therefore essential.

Power Restoration Should Be Sequenced

After a major outage, restoring everything simultaneously can cause excessive inrush and destabilize the emergency source.



Load Shedding

When emergency capacity is limited, noncritical loads can be automatically disconnected so that essential clinical services remain supplied.

N+1 Redundancy

Definition

N is the capacity required to meet the design load. N+1 adds one additional unit so the system can tolerate one unit being unavailable.

Example:

- ▶ design requires two generators operating;
- ▶ three equally sized generators are installed;
- ▶ one can be maintained or fail while two remain available.

Caution

Redundancy is useful only if common-mode failures are controlled. Three generators connected through one vulnerable switchboard may still share a single point of failure.

Single Points of Failure

Potential single points of failure include:

- ▶ one transformer supplying the entire critical system;
- ▶ one ATS feeding multiple high-risk departments;
- ▶ one cable route exposed to the same fire or flood zone;
- ▶ one cooling system serving all UPS modules;
- ▶ one fuel pump serving all generators;
- ▶ one communication network for every electrical alarm.

Resilience Question

“What one failure could still take down the whole system?” is one of the most valuable questions in hospital electrical design.

Earthing: Why Is It Important?

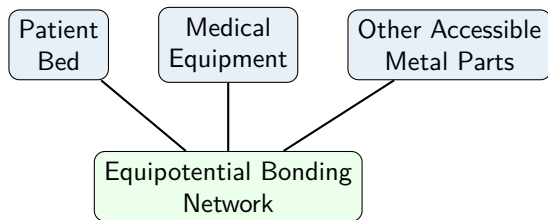
Earthing provides a reference and fault-current path that supports protective-device operation and reduces touch-voltage risk.

- ▶ exposed conductive parts should not remain dangerously energized after a fault;
- ▶ protective conductors must remain continuous;
- ▶ earth connections require inspection and testing;
- ▶ earthing and bonding must be coordinated with the adopted electrical system;
- ▶ sensitive medical locations require special attention to potential differences.

Earthing is a safety system, not merely a wire connected to ground.

Equipotential Bonding in Patient Areas

The goal of equipotential bonding is to reduce dangerous voltage differences between accessible conductive parts.



Clinical Relevance

A patient connected to multiple devices may simultaneously contact several conductive objects; minimizing potential differences reduces shock risk.

Electrical Shock: Macroshock and Microshock

Macroshock

Current passes through the body from external contact. Risk depends on current magnitude, path, duration, skin condition and other factors.

Microshock Concept

When an invasive conductive pathway reaches the heart, very small currents can become clinically significant. This motivates stricter control of leakage and potential differences in certain patient environments.

Biomedical engineers must understand the patient connection, not just the equipment enclosure.

Leakage Current

Leakage current is unintended current that flows through insulation, protective earth or patient-connected pathways.

Possible contributors include:

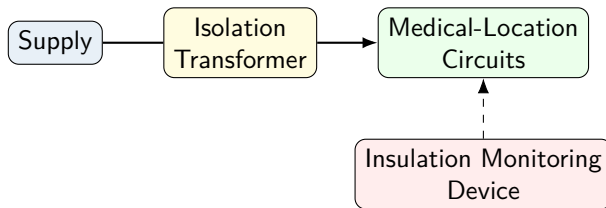
- ▶ capacitive coupling in power supplies;
- ▶ damaged insulation;
- ▶ incorrect wiring or protective-earth failure;
- ▶ multiple interconnected devices;
- ▶ fluid contamination or aging components.

Safety Testing

Medical electrical equipment is tested according to applicable product and in-service safety standards. Acceptable limits depend on device classification and test condition; use the current standard and manufacturer requirements.

Isolated Power Systems in Medical Locations

Some high-risk medical locations may use a medical IT / isolated power arrangement according to the adopted standard.



Purpose

Maintain continuity on the first insulation fault while immediately alarming staff so the fault can be located and corrected.

Insulation Monitoring: Teaching Example

Suppose an isolated medical circuit develops its first insulation fault.

1. the insulation monitoring device detects degradation;
2. an audible / visual alarm is generated;
3. the circuit may remain energized according to the design intent;
4. engineering staff identify and remove the faulty device or circuit;
5. the fault is documented and corrective action verified.

Why Not Ignore the Alarm?

The first fault may not interrupt service, but a second fault can remove the safety advantage and create a dangerous condition.

Protective Devices

Hospital electrical systems use protective devices such as:

- ▶ fuses and circuit breakers;
- ▶ residual-current protective devices where appropriate;
- ▶ earth-fault protection;
- ▶ surge-protective devices;
- ▶ over/undervoltage and frequency protection;
- ▶ generator and transformer protection;
- ▶ arc-fault mitigation strategies where required by the adopted design.

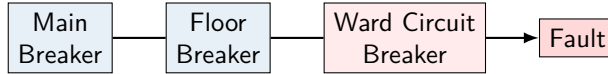
Protection Must Balance Two Goals

Safety: clear dangerous faults quickly.
healthy critical circuits.

Continuity: avoid unnecessary shutdown of

Selective Coordination

Consider a fault on one ward circuit.



Desired Behaviour

The protective device nearest the fault should operate first whenever coordination permits, leaving unrelated critical circuits energized.

Poor Coordination

If the main breaker trips for a local ward fault, the electrical design has created an unnecessarily large clinical disturbance.

Socket Outlets in Patient-Care Areas

Engineering considerations include:

- ▶ sufficient quantity to avoid unsafe extension boards;
- ▶ correct circuit and source identification;
- ▶ segregation of normal and essential circuits where applicable;
- ▶ robust protective-earth continuity;
- ▶ mechanical integrity and secure connections;
- ▶ location that avoids trip hazards and fluid exposure;
- ▶ planned inspection and testing.

Practical Point

A well-designed ICU often needs many more outlets than originally expected because each bed accumulates monitors, pumps, ventilators, warming devices, chargers and IT equipment.

Extension Boards: A Hidden Hospital Hazard

Uncontrolled extension boards and multi-plug adapters can create:

- ▶ circuit overload;
- ▶ poor earth continuity;
- ▶ loose connections and heating;
- ▶ fluid-ingress risk;
- ▶ trip hazards;
- ▶ accidental disconnection;
- ▶ difficulty tracing which source actually powers a device.

Engineering Solution

Provide adequate fixed outlets, enforce an approved-device policy, and treat outlet availability as part of clinical-area planning.

Power Quality in Hospitals

Sensitive medical equipment can be affected even when the supply has not completely failed. Common disturbances include:

- ▶ voltage sag and swell;
- ▶ short interruptions;
- ▶ harmonics;
- ▶ transients and surges;
- ▶ frequency deviation;
- ▶ unbalance;
- ▶ electrical noise and poor grounding.

Symptoms

Unexpected reboot, image artifacts, communication errors, nuisance alarms, premature component failure and unexplained device resets.

Harmonics and Nonlinear Loads

Modern hospitals contain many nonlinear electronic loads:

- ▶ switched-mode power supplies;
- ▶ LED lighting drivers;
- ▶ UPS systems;
- ▶ variable-frequency drives;
- ▶ imaging and laboratory equipment;
- ▶ servers and IT systems.

Why Harmonics Matter

They can increase neutral current, heating and losses; distort voltage; interact with capacitors; and reduce power-quality margin.

Measure first. Mitigation should be based on power-quality data, not assumptions.

Power Factor

For an AC load:

$$\text{Power Factor} = \frac{P \text{ (kW)}}{S \text{ (kVA)}}$$

A low power factor can increase current and infrastructure loading for the same useful power.

Example

A 400 kW load at PF 0.80 requires:

$$S = \frac{400}{0.80} = 500 \text{ kVA}$$

At PF 0.95:

$$S = \frac{400}{0.95} \approx 421 \text{ kVA}$$

Caution

Power-factor correction must be engineered carefully in systems with harmonics and generators.

Surge and Lightning Protection

Hospitals contain expensive and safety-critical electronic systems that can be damaged by transient overvoltages.

A coordinated strategy may include:

- ▶ external lightning-protection system where required;
- ▶ equipotential bonding;
- ▶ surge-protective devices at appropriate distribution levels;
- ▶ good cable routing and earthing;
- ▶ protection of data and communication interfaces;
- ▶ inspection after major events.

Key Idea

One small plug-in surge protector is not a complete lightning and surge-protection system.

Electromagnetic Compatibility (EMC)

Electrical systems and medical devices share the same environment with wireless and high-frequency sources.

Potential problems:

- ▶ interference between devices;
- ▶ noise coupling through power or signal cables;
- ▶ poor cable segregation;
- ▶ malfunction near powerful transmitters;
- ▶ grounding loops and communication disturbances.

Biomedical Engineer's Role

Investigate whether an apparent device failure is truly a device fault or a power / EMC environment problem.

Lighting Systems in Hospitals

Lighting design must support:

- ▶ clinical examination and procedures;
- ▶ patient comfort and circadian needs;
- ▶ safe movement and wayfinding;
- ▶ task lighting at nursing and medication areas;
- ▶ emergency evacuation;
- ▶ infection-control-compatible fixtures where required;
- ▶ energy efficiency without compromising clinical performance.

Emergency Lighting

Emergency illumination must be considered separately from normal convenience lighting because its purpose is safe continuity and egress.

Electrical Design of an ICU Bed Space

A single ICU bed may need electrical support for:

- ▶ ventilator;
- ▶ patient monitor;
- ▶ multiple infusion / syringe pumps;
- ▶ suction and accessories;
- ▶ feeding pump;
- ▶ warming / cooling devices;
- ▶ portable imaging or diagnostic equipment;
- ▶ computers and communication devices.

Planning Implication

Outlet count, circuit diversity, emergency-source availability and cable management should be determined from the clinical workflow, not from a generic room template.

Electrical Design of an Operation Theatre

An OT requires coordinated power for:

- ▶ anaesthesia workstation and monitors;
- ▶ surgical lights;
- ▶ electrosurgical unit;
- ▶ endoscopy / laparoscopy towers;
- ▶ pumps, warming devices and powered tables;
- ▶ imaging systems where used;
- ▶ HVAC and controls;
- ▶ emergency and communication systems.

Design Principle

Power continuity, shock protection, bonding, cable management and device compatibility must be planned as one integrated OT safety system.

Electrical Requirements of Imaging Equipment

Large imaging systems may impose special requirements:

- ▶ high instantaneous power demand;
- ▶ dedicated feeders;
- ▶ voltage regulation and power quality;
- ▶ harmonic considerations;
- ▶ controlled shutdown sequence;
- ▶ dedicated UPS for control / computing subsystems;
- ▶ cooling and HVAC interdependence;
- ▶ manufacturer-specified earthing and installation conditions.

Commissioning Lesson

Never install a high-end imaging system before verifying the electrical infrastructure against the manufacturer's site-planning data.

Power and Medical Equipment: Shared Responsibility

Electrical / Facility Team

- ▶ source and distribution;
- ▶ protection and earthing;
- ▶ generator and UPS;
- ▶ switchgear maintenance;
- ▶ power-quality monitoring.

Biomedical / Clinical Engineering

- ▶ equipment electrical-safety tests;
- ▶ device power requirements;
- ▶ battery health;
- ▶ failure investigation;
- ▶ commissioning interface.

Safe patient care requires both teams to communicate.

Generator Fuel Is Part of Electrical Resilience

Backup duration depends not only on generator rating but also on fuel logistics.

Questions to ask:

- ▶ How long can critical services operate on stored fuel?
- ▶ Is the fuel quality monitored?
- ▶ Are transfer pumps and controls redundant?
- ▶ Can fuel be replenished during flood, storm or civil disruption?
- ▶ Is there a safe manual backup procedure?
- ▶ Are fuel leaks, fire risk and environmental requirements controlled?

Disaster Planning

A generator without assured fuel availability is not a complete emergency-power plan.

Electrical Rooms: Design and Access

Electrical rooms should support safe operation and maintenance.

Key considerations:

- ▶ restricted authorized access;
- ▶ adequate working clearance;
- ▶ environmental control and ventilation;
- ▶ protection from water leakage and flooding;
- ▶ suitable fire detection / protection;
- ▶ clear labeling and updated diagrams;
- ▶ housekeeping and removal of combustible storage;
- ▶ safe escape and emergency access.

Water and Electricity: A Dangerous Combination

Common hospital water-related electrical risks include:

- ▶ plumbing leakage above switchboards;
- ▶ flood-prone basement substations;
- ▶ wet housekeeping near energized outlets;
- ▶ leaking dialysis or laboratory water systems;
- ▶ condensation from HVAC systems;
- ▶ roof leakage into electrical risers.

Resilience Principle

Location is a protective measure. Critical electrical equipment should not be placed where foreseeable water ingress can disable it.

Maintenance of Hospital Electrical Systems

A planned electrical-maintenance programme may include:

- ▶ visual inspection and housekeeping;
- ▶ torque / connection checks where appropriate;
- ▶ insulation and earth-continuity testing;
- ▶ protective-device testing;
- ▶ thermal imaging of loaded connections;
- ▶ generator start and load testing;
- ▶ ATS functional tests;
- ▶ UPS and battery inspection / capacity testing;
- ▶ alarm and monitoring verification;
- ▶ review of trip and failure history.

Thermography as Predictive Maintenance

Infrared thermography can identify abnormal heating before visible failure.

Potential causes of hot spots:

- ▶ loose connections;
- ▶ overloaded conductors;
- ▶ unbalanced phases;
- ▶ deteriorating contacts;
- ▶ faulty breakers or bus joints.

Important

Thermal images must be interpreted with load level, emissivity, access safety and equipment condition in mind. A colourful image alone is not a diagnosis.

Generator Testing: What Should Be Verified?

- ▶ automatic start on simulated source failure;
- ▶ starting-battery condition;
- ▶ stable voltage and frequency;
- ▶ cooling and lubrication parameters;
- ▶ fuel system operation;
- ▶ alarms and shutdown protections;
- ▶ ATS transfer / retransfer;
- ▶ ability to carry representative load;
- ▶ recovery after step-load changes;
- ▶ documentation of every test.

Testing Philosophy

Test the **system** from failure detection to load restoration, not only the engine.

UPS Maintenance

A UPS can become a single point of failure if neglected.

Maintenance includes:

- ▶ alarm-history review;
- ▶ fan and cooling inspection;
- ▶ capacitor and power-module condition;
- ▶ bypass verification;
- ▶ battery voltage, temperature and impedance / health checks;
- ▶ autonomy or capacity testing as appropriate;
- ▶ cleaning and environmental control;
- ▶ firmware / control review under change control;
- ▶ documented replacement planning.

Electrical Safety Before Maintenance

Before working on electrical equipment:

- ▶ identify all energy sources;
- ▶ plan the clinical impact of isolation;
- ▶ obtain required authorization;
- ▶ isolate and secure the source;
- ▶ verify absence of voltage using appropriate test equipment;
- ▶ control stored electrical / mechanical energy;
- ▶ use suitable PPE and tools;
- ▶ communicate restoration before re-energization.

Hospital-Specific Challenge

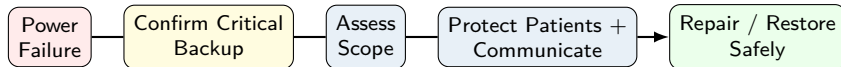
The technical safest shutdown may still be clinically unsafe if a critical service has no alternate supply. Maintenance planning must therefore involve clinical users.

Planned Electrical Shutdown

A planned shutdown should include:

1. scope and risk assessment;
2. affected departments and circuits;
3. backup-source verification;
4. temporary power where required;
5. clinical approval and communication;
6. shutdown / isolation sequence;
7. work execution and testing;
8. controlled restoration;
9. post-restoration verification;
10. incident / lesson documentation.

Emergency Outage Response



During an Outage

Engineering teams should prioritize clinical consequence, not merely the chronological order of fault calls.

Case Study 1: ICU Reboots During Generator Transfer

During a grid outage, the generator starts correctly. However, several bedside monitors reboot during the transfer.

Possible Causes

- ▶ monitors not connected to UPS-backed outlets;
- ▶ depleted internal batteries;
- ▶ UPS overload or bypass condition;
- ▶ excessive transfer interruption;
- ▶ poor outlet labeling / connection error.

Investigation

Do not blame the monitors immediately. Trace the full power path from source to outlet to device battery.

Case Study 2: Repeated Breaker Trip in OT

An OT circuit breaker trips intermittently during procedures.

Ask:

1. What devices are connected when the trip occurs?
2. Has the clinical load increased since original design?
3. Is there a faulty device or cable?
4. Is the breaker correctly rated and healthy?
5. Are leakage or harmonic effects involved?
6. Has an unauthorized extension board been added?
7. Are downstream circuits correctly segregated?

Lesson

Repeated tripping is a symptom. Resetting the breaker without root-cause analysis is not maintenance.

Case Study 3: Generator Starts but ICU Has No Power

A grid failure occurs. The generator starts and appears healthy, but the ICU remains without essential power.

Potential failure points:

- ▶ ATS fails to transfer;
- ▶ control power is missing;
- ▶ downstream breaker is open;
- ▶ essential feeder cable has failed;
- ▶ wrong circuit classification / labeling;
- ▶ load-shedding logic removed the wrong feeder;
- ▶ local distribution-board fault.

Backup generation and backup distribution are equally important.

Case Study 4: UPS Says “Healthy” but Battery Fails

A UPS display shows normal operation for months. During an outage, runtime is only two minutes instead of the expected fifteen.

Likely Questions

- ▶ Were batteries beyond expected service life?
- ▶ Was ambient temperature excessive?
- ▶ Was actual load higher than design?
- ▶ Were capacity tests performed?
- ▶ Did one weak battery string limit the bank?
- ▶ Was battery monitoring calibrated and reviewed?

Lesson

“No alarm” is not proof of stored-energy capacity.

Power Monitoring and Building Management Systems

Electrical monitoring can include:

- ▶ voltage, current, frequency and power factor;
- ▶ energy consumption;
- ▶ breaker status;
- ▶ generator status and fuel level;
- ▶ UPS load and battery alarms;
- ▶ power-quality events;
- ▶ transformer temperature;
- ▶ ATS status;
- ▶ critical distribution alarms.

From Data to Action

Monitoring is valuable only when alarms are prioritized, routed to responsible staff and linked to an escalation procedure.

Electrical System KPIs

Useful indicators may include:

- ▶ number and duration of unplanned power interruptions;
- ▶ generator start success rate;
- ▶ ATS transfer-test compliance;
- ▶ UPS battery-capacity compliance;
- ▶ critical electrical PM compliance;
- ▶ power-quality event count;
- ▶ repeated breaker trips;
- ▶ electrical incident / near-miss rate;
- ▶ percentage of critical alarms acknowledged within target time.

Management Principle

Do not measure only energy consumption. Measure **continuity and safety performance** too.

Energy Efficiency Without Compromising Safety

Opportunities include:

- ▶ high-efficiency transformers and motors;
- ▶ LED lighting and intelligent controls;
- ▶ variable-speed drives for suitable pumps and fans;
- ▶ power-factor and power-quality improvement;
- ▶ submetering and energy dashboards;
- ▶ optimized HVAC scheduling outside critical areas;
- ▶ peak-demand management;
- ▶ renewable generation where technically and economically suitable.

Priority

Energy savings must never disable required redundancy, ventilation, lighting, infection-control or emergency functions.

Solar PV and Hospitals

Solar PV can reduce grid energy consumption, but by itself it does not automatically provide emergency power.

A resilient design may need:

- ▶ suitable inverter architecture;
- ▶ intentional islanding capability if permitted and designed;
- ▶ battery energy storage where required;
- ▶ protection and anti-islanding coordination;
- ▶ generator / microgrid control strategy;
- ▶ clear priority of critical loads.

Key Message

Renewable energy and **emergency power** are related but not identical design objectives.

Electrical Fire Risk

Electrical fires may originate from:

- ▶ overloaded circuits;
- ▶ loose connections;
- ▶ damaged cables;
- ▶ inappropriate extension boards;
- ▶ poor maintenance;
- ▶ battery systems;
- ▶ switchgear faults;
- ▶ contaminated or overheated equipment.

Prevention

Correct design, protection, inspection, thermal monitoring, housekeeping and disciplined maintenance reduce risk.

Electrical Incident Investigation

After an incident, preserve evidence and reconstruct the event.

Collect:

- ▶ time-stamped alarms and BMS logs;
- ▶ breaker / relay event records;
- ▶ UPS and generator logs;
- ▶ connected-load list;
- ▶ maintenance history;
- ▶ recent modifications;
- ▶ staff observations;
- ▶ relevant waveforms or power-quality recordings;
- ▶ photographs and failed components where safe.

Goal

Find the system cause and prevent recurrence, rather than merely identify who last touched the equipment.

Electrical Change Control

Hospitals continuously add new equipment. Every major addition should trigger an engineering review.

Ask:

- ▶ Is spare electrical capacity available?
- ▶ Does the equipment need normal or emergency supply?
- ▶ Is a dedicated circuit required?
- ▶ What are inrush, harmonic and power-quality requirements?
- ▶ Is earthing / bonding suitable?
- ▶ Does the UPS or generator need resizing?
- ▶ Are drawings and asset records updated?

Rule

Procurement of medical equipment is also an infrastructure decision.

Commissioning a New Critical Electrical System

Commissioning should verify not just installation, but performance under realistic conditions.

1. document design intent;
2. inspect installation and labeling;
3. verify protective settings and interlocks;
4. test generator and ATS sequence;
5. test UPS / bypass / battery functions;
6. simulate selected failure scenarios;
7. verify alarms and monitoring;
8. train operations staff;
9. hand over updated drawings and test records.

Electrical Safety Culture

A safe electrical system requires more than equipment.

- ▶ clear responsibilities;
- ▶ competency-based authorization;
- ▶ accurate labeling and documentation;
- ▶ formal permit / isolation procedures;
- ▶ incident and near-miss reporting;
- ▶ planned drills for major power loss;
- ▶ cross-training of engineering and clinical teams;
- ▶ routine review of critical failure scenarios.

Resilience is engineered, tested and rehearsed.

What Should a Biomedical Engineer Check Before Installing Equipment?

1. rated voltage, frequency and maximum input power;
2. plug, socket and protective-earth compatibility;
3. normal vs essential / UPS supply requirement;
4. leakage-current and electrical-safety classification;
5. manufacturer earthing / isolation requirements;
6. inrush and UPS compatibility;
7. environmental and cooling requirements;
8. network / data / EMC considerations;
9. installation acceptance and baseline safety tests.

Hospital Electrical Risk Assessment

A simple risk model can consider:

$$\text{Risk} \propto \text{Probability of Failure} \times \text{Clinical Consequence} \times \text{Exposure}$$

Examples of High-Risk Assets

- ▶ critical ICU distribution board;
- ▶ OT isolation transformer / monitoring system;
- ▶ emergency generator control system;
- ▶ main ATS;
- ▶ UPS serving life-support or core network infrastructure.

Risk ranking helps determine inspection, redundancy, spares and testing intensity.

FMEA Example: Emergency Power System

Failure Mode	Effect	Possible Cause	Control
Generator fails to start	Essential loads remain unsupported	Battery / starter / fuel fault	Scheduled start test, battery monitoring
ATS fails to transfer	Generator healthy but load remains dead	Control or mechanical failure	Functional transfer testing
UPS battery degraded	Short autonomy	Aging / heat / failed cells	Capacity and condition testing
Critical breaker trips	Local service interruption	Overload / fault / poor coordination	Load review, protection study

Use of FMEA

FMEA turns “What if?” questions into planned engineering controls before a real clinical emergency occurs.

Standards and Regulatory Framework

Hospital electrical design should follow the current editions adopted by the project / institution.

Examples include:

- ▶ Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2023, India;
- ▶ IEC 60364-7-710:2021 for electrical installations in medical locations;
- ▶ IEC 60601 series for medical electrical equipment;
- ▶ NABH Hospital Accreditation Standards, 6th Edition, Facility Management and Safety requirements;
- ▶ applicable National Building Code, fire, local electrical-inspector and utility requirements;
- ▶ NFPA 99 where used by the institution or project specification.

Important

This lecture explains engineering principles. Final design values, transfer times, testing intervals and protection requirements must come from the legally adopted current code and project specifications.

Classroom Activity: Classify the Loads

Classify each load as requiring no-break / near-no-break support, rapid emergency restoration, or normal supply only. Justify your answer.

1. ICU ventilator;
2. waiting-area television;
3. OT surgical light;
4. central server supporting patient records;
5. cafeteria refrigerator;
6. fire alarm panel;
7. MRI control computer;
8. office printer;
9. medical gas alarm panel;
10. general corridor lighting.

Discussion

Different institutions may classify some loads differently. The important part is the **clinical reasoning**.

Classroom Numerical: UPS Planning

A critical care area has the following selected no-break loads:

- ▶ monitors: 4 kW;
- ▶ ventilators: 8 kW;
- ▶ infusion systems: 2 kW;
- ▶ IT / communication: 6 kW.

Assume $PF = 0.90$ and 30% capacity margin.

Calculate

1. total kW;
2. minimum apparent-power requirement in kVA;
3. planning kVA after the 30% margin;
4. ideal battery energy for 20 minutes of full-load autonomy.

Classroom Design Activity: ICU Electrical Plan

In groups, design the electrical concept for a six-bed ICU.

Include:

- ▶ normal and essential power paths;
- ▶ UPS-backed loads;
- ▶ approximate outlet strategy per bed;
- ▶ backup generator interface;
- ▶ earthing and bonding concept;
- ▶ alarm / monitoring points;
- ▶ one single-point-of-failure risk;
- ▶ one maintenance / testing requirement.

Deliverable

Draw a one-page block diagram and explain the design in three minutes.

Mini-Project Ideas

Possible B.Tech Biomedical Engineering projects:

- ▶ IoT dashboard for generator, ATS and UPS status;
- ▶ critical-load classification tool for hospital departments;
- ▶ battery state-of-health monitoring for UPS systems;
- ▶ power-quality monitoring at ICU / laboratory outlets;
- ▶ anomaly detection from electrical alarm logs;
- ▶ QR-based electrical PM and inspection records;
- ▶ energy-and-reliability dashboard for a hospital block;
- ▶ simulation of emergency load shedding and restoration;
- ▶ risk-based FMEA of a hospital emergency-power system.

Key Engineering Questions

When evaluating any hospital electrical system, ask:

1. What happens if normal power disappears now?
2. Which loads must never lose power?
3. What is the alternate source and how is it transferred?
4. What single failure could disable the backup system?
5. Are earthing, bonding and protection suitable for medical locations?
6. Can the system handle future medical equipment?
7. Are generator, ATS, UPS and batteries tested under realistic conditions?
8. Can staff identify and respond to alarms quickly?

Summary

- ▶ Hospital electrical systems are clinical safety systems as well as engineering utilities.
- ▶ Load criticality determines continuity, redundancy and backup requirements.
- ▶ Generators, ATS systems, UPS and batteries provide complementary layers of resilience.
- ▶ Earthing, equipotential bonding, leakage-current control and protection are central to patient safety.
- ▶ Power quality can affect sensitive medical equipment even without a total outage.
- ▶ Planned testing, maintenance, monitoring and outage drills are essential.
- ▶ Biomedical and electrical engineers must jointly evaluate the power needs of medical technology.

A hospital power system must fail safely, recover predictably and protect the patient throughout.

Assignment for the Next Lecture

Prepare a one-page electrical-resilience plan for one hospital area:

- ▶ ICU, operation theatre, emergency department, laboratory or radiology.

Include:

1. five major electrical loads;
2. load criticality classification;
3. normal and emergency source;
4. which loads require UPS backup;
5. one earthing / shock-safety consideration;
6. one power-quality risk;
7. one maintenance / test activity;
8. one possible single point of failure.

Submission Format

One page with a simple electrical block diagram.

Suggested Reading

- ▶ Central Electricity Authority, *Measures relating to Safety and Electric Supply Regulations, 2023*, Government of India.
- ▶ IEC 60364-7-710:2021, *Low-voltage electrical installations - Medical locations*.
- ▶ IEC 60601 series, *Medical electrical equipment - General and particular safety requirements*.
- ▶ National Accreditation Board for Hospitals and Healthcare Providers, *Hospital Accreditation Standards, 6th Edition - Facility Management and Safety*.
- ▶ NFPA 99, *Health Care Facilities Code*, where applicable to the project or institution.
- ▶ G. D. Kunders, *Hospitals: Facilities Planning and Management*.
- ▶ Manufacturer site-planning guides for generators, UPS systems, imaging systems and critical medical equipment.

Next Lecture: HVAC and Air-Conditioning Systems in Hospitals

Thank You

Questions and Discussion